Chapter 1
Introduction: Derivatives and Their Markets

A derivative is a contract whose value depends on something else. That is nearly the whole definition, and its emptiness is the point: the something else can be a commodity price, an interest rate, an exchange rate, a share price, a credit event, an index of volatility, a quantity of rainfall, or the price of another derivative. What unites the instruments in this book is not what they reference but what they do—they separate exposure to a risk from ownership of the thing that carries it.

That separation is the entire economic content of the subject. A farmer who wants the price of wheat fixed in September need not sell the wheat in March. A pension fund that wants thirty-year interest rate exposure need not buy thirty-year bonds. A bank that has made a loan it wants to keep, to a borrower whose credit it no longer wants, need not sell the loan. In each case a contract does what a transaction in the underlying asset would otherwise have done, usually with less capital, less disruption, and more precision—and, as the rest of this book documents at length, with risks of its own that are easy to underestimate precisely because so little cash changes hands at the outset.

This chapter is the map; every claim in it is developed properly later.

Sections 1.1 to 1.3 set out the four instruments, where they came from, and why they exist. Sections 1.4 to 1.7 describe the markets: exchange-traded against over-the-counter, the legal architecture of master agreements and netting, how large the market is once gross notional is read correctly, and what can serve as an underlying. Sections 1.8 to 1.10 are the participants: who uses derivatives and for what, why the labels hedger, speculator, and arbitrageur blur, and the leverage that makes every position larger than its cash. Sections 1.11 and 1.12 record what has gone wrong and how regulation responded after 2008, and Section 1.13 explains how the chapters fit together.

1.1 THE FOUR BUILDING BLOCKS

Almost every derivative in existence is a combination of four contracts. Learning the four well is more useful than memorizing the hundreds of named structures assembled from them.

A forward obliges one party to buy, and the other to sell, a specified asset on a specified future date at a price agreed now, the **forward price**. No money changes hands at inception; the forward price is set so that the contract is worth zero to both sides on day one. It is the simplest derivative and the subject of Chapter 5.

A futures contract is a forward that has been standardized, listed on an exchange, and settled daily in cash; the price at which it trades is the **futures price**. Daily settlement is more than a technicality: it converts a single payment at maturity into a stream of payments along the way, so that whichever side the price moves against must find cash now rather than at maturity, and Chapters 2 and 8 are largely about the consequences.

A swap exchanges two streams of payments (most commonly a fixed interest rate for a floating one) over a series of dates. A swap is economically a portfolio of forwards, one per payment date, bundled into a single contract with a single price. Chapter 7 develops them.

An option gives its holder the *right*, not the obligation, to buy (a call) or sell (a put) at a set price, the strike price. Because the holder will exercise only when it pays to do so, the option can never be worth less than zero to them, and they must therefore pay for it at the outset with a **premium**. Part II is devoted to options.

The essential distinction is between the first three and the last. Forwards, futures, and swaps are *symmetric*: both parties are obliged, gains and losses run in both directions, and the **payoff**, what the contract delivers at maturity, is a straight line. Options are *asymmetric*: one party has a choice, the payoff is kinked, and the premium is the price of the kink. Figure 1.1 shows the difference in terms of **profit**, which is the payoff net of any premium paid or received.

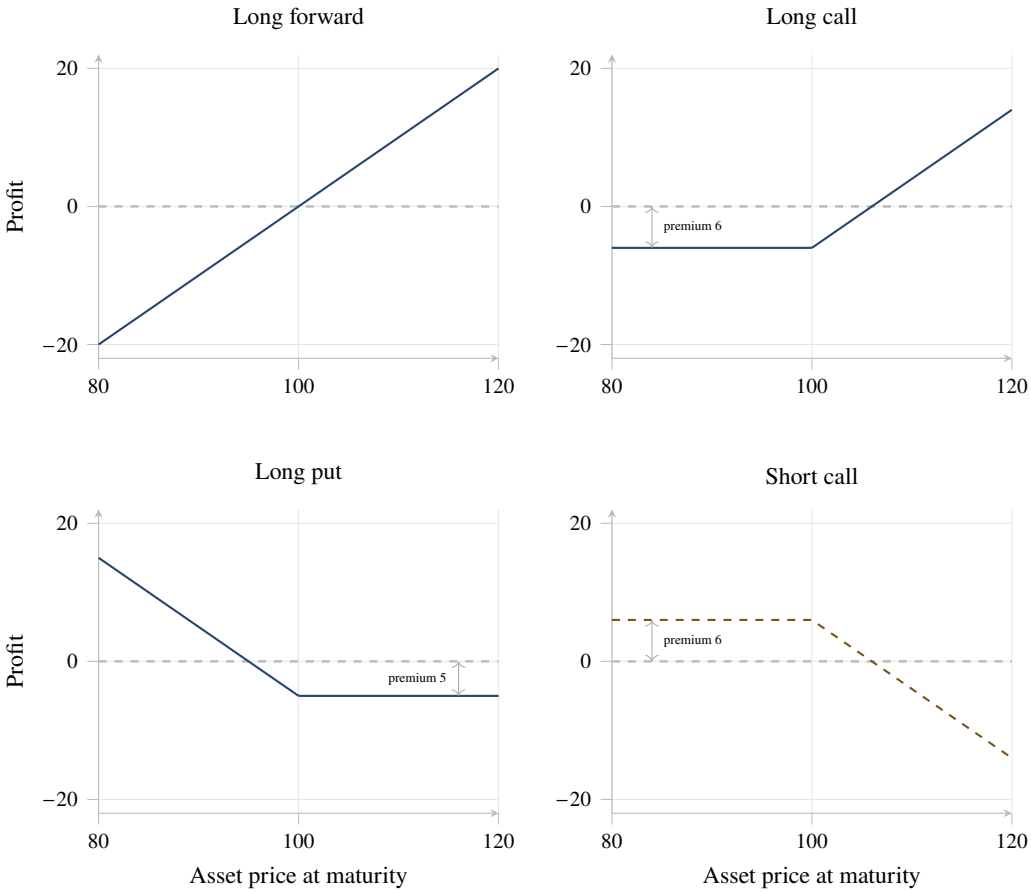


Figure 1.1: Profit at maturity, premium included. Strike and forward price 100; call premium 6, put premium 5.

Three features of Figure 1.1 are worth fixing now. The forward's line passes through zero at the forward price, so the contract starts life worth nothing and can end worth a great deal in either direction. The long call's worst case is bounded at the premium, however far the market falls. And the short call, in the fourth panel, has the mirror-image profile: a gain capped at the premium and an unbounded loss. That asymmetry is why option writing is a business for firms that can hedge and post collateral, and why it recurs in almost every chapter about things going

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wrong.

1.2 WHERE THESE MARKETS CAME FROM

Derivatives are often presented as a modern invention, and the impression is wrong in a way that obscures how the market works. Contracts for future delivery are ancient; what is modern is the standardization, the clearing, and the range of risks that can be referenced. More useful than a chronology is the pattern the chronology reveals: each major instrument appeared shortly after a risk became intolerable and a way of measuring it became available.

The Dojima rice market in Osaka, formally sanctioned in 1730, traded standardized claims on rice for future delivery, with a clearing house and margin: the essential apparatus of Chapter 2, two centuries early. The Chicago Board of Trade, founded in 1848, standardized grain contracts in the 1860s for the same reason: buyers and sellers separated by a season and a railroad needed a price agreed in advance.

The modern market dates from the breakdown of the Bretton Woods system. When fixed exchange rates ended in the early 1970s, currency risk became a problem every multinational suddenly had, and in May 1972 the Chicago Mercantile Exchange listed the first financial futures, on foreign currencies. Inflation and interest rate volatility followed, and with them the first interest rate futures in 1975–77 and the Eurodollar contract in 1981, on three-month US dollar deposits held at banks outside the United States—the first futures contract settled purely in cash because its underlying could not be delivered at all.

Options followed the same logic on a different track. They had traded over the counter for centuries, illiquidly and with unreliable pricing. The Chicago Board Options Exchange opened in April 1973 with listed calls on sixteen stocks; the Black–Scholes–Merton papers appeared the same year. It is tempting to say the model created the market. It is more accurate to say that standardization and a workable valuation method arrived together, and that neither would have sufficed alone, a point Chapter 15 returns to.

Swaps arrived from a different direction: not an exchange, but a single negotiated trade. The 1981 currency swap between IBM and the World Bank, arranged by Salomon Brothers, is conventionally cited as the first. Within a decade the swap market was larger than the exchange-traded market, entirely bilateral and documented under agreements that had to be invented as it grew (Section 1.5). Credit default swaps, which pay the buyer if a specified borrower defaults, emerged from the same bilateral tradition in the mid-1990s, developed by banks that wanted to keep lending to clients to whom they already had too much credit exposure.

The last two decades reverse the pattern. Their innovations have been structural, not new instruments: central clearing extended to over-the-counter (OTC) contracts after 2009, mandatory margining completed in 2022, and the replacement of the London Interbank Offered Rate (LIBOR) by overnight risk-free rates between 2021 and 2023—a change of foundations rather than of instruments. Newer underlyings continue to appear, and Section 1.7 surveys them, but the striking feature of the modern era is that the four building blocks of Section 1.1 have not changed. What changed is who stands between the two parties, and what they must post.

1.3 WHY DERIVATIVES EXIST

Why anyone bothers is a fair question, because the alternatives are real and sometimes better. A firm exposed to a price it dislikes has several choices that do not involve derivatives. It can hold physical inventory, which costs storage and capital. It can buy insurance, which works for events that are verifiable and idiosyncratic but not for market prices, since an insurer cannot diversify away a move that hits every policyholder at once. It can restructure operations (an airline can hedge fuel by owning a refinery, as Delta did in 2012), which works but ties up capital and management attention in a business the firm did not choose to be in. It can hold

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more equity, absorbing the risk rather than transferring it. Or it can simply accept the exposure, which for a diversified shareholder may be the correct answer and is discussed in Section 3.2.

Derivatives compete with these on three grounds:

Precision. A contract can be written on exactly the risk in question (one currency pair, one maturity, one delivery month) rather than on a bundle that happens to contain it.

Capital efficiency. Transferring $100 million of exposure requires posting margin measured in single-digit millions rather than committing $100 million. This is a genuine advantage and simultaneously the source of most of the disasters in Section 1.11; Section 1.10 treats it as one topic rather than two.

Reversibility. A hedge can be lifted when the exposure changes; the sale of a business cannot.

Against these sits the standing objection: because so little cash is required at inception, the size of a derivatives position is not disciplined by the size of the balance sheet behind it. Every episode in Section 1.11 is a variation on that sentence.

1.4 EXCHANGE-TRADED AND OVER-THE-COUNTER MARKETS

Derivatives trade in two structurally different places, and the difference used to be sharp. It is now more subtle, because regulation has pushed the two toward each other.

Exchange-traded contracts are standardized: the exchange fixes the size, the underlying, the delivery months, and the settlement terms, and participants choose only price and quantity. Every trade is novated to a central counterparty (CCP), which replaces it with two contracts, one between each party and the CCP, so the CCP becomes buyer to every seller and seller to every buyer, and no participant bears the credit risk of another. Standardization concentrates liquidity (any two participants wanting the same contract meet in the same order book) at the cost of fit: the contract may not match the exposure a hedger actually has, which is the basis risk of Chapter 3.

Over-the-counter contracts are negotiated bilaterally between two institutions and can specify anything the parties agree on: a non-standard maturity, an unusual underlying, a bespoke payoff. The fit is exact. Historically the price of that fit was credit risk: each party depended on the other's solvency for the life of the contract, which for a thirty-year swap is a long time. Figure 1.2 shows the two, and the hybrid that now dominates.

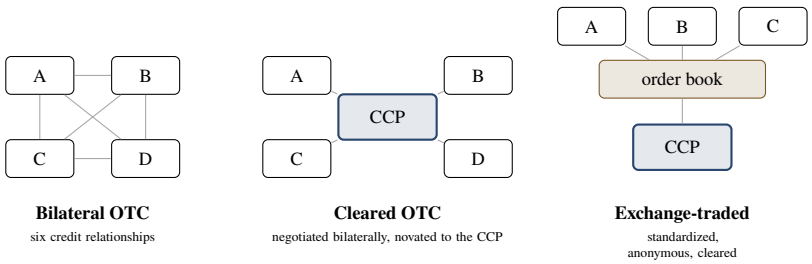


Figure 1.2: How a trade reaches its counterparty. Four participants shown.

The middle panel is where most of the market now sits. Following the G20's 2009 commitments (Section 1.12), standardized OTC contracts, above all interest rate swaps, must be cleared through a central counterparty. Such a trade is negotiated bilaterally, like an OTC contract, and then novated to a CCP, like a futures trade. It inherits the flexibility of one structure and the credit mechanics of the other, and that is why the old question "exchange-traded or OTC?" now has three answers rather than two. Chapter 2 works through what a CCP actually does with the risk it absorbs, which is not the same as making the risk disappear.

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1.5 THE LEGAL ARCHITECTURE: MASTER AGREEMENTS AND NETTING

A bilateral derivative is a promise extending years into the future, and what makes such a promise financeable is not the pricing model but the document. This is not a legal aside. The reduction from $23 trillion to $3.4 trillion in the next section is entirely an artifact of the arrangement described here, and Chapters 8 and 9 assume it throughout.

Two institutions did not exist when the swap market began and had to be invented as it grew. The first is the contract published by the International Swaps and Derivatives Association (ISDA), the **ISDA Master Agreement**, now the standard for essentially all bilateral derivatives. Its central provision is deceptively simple: all transactions between the two parties, however many and however different, form a **single agreement**. A party cannot default on one swap and perform on another.

That provision exists to enable **close-out netting**. If one party defaults, every outstanding transaction between them terminates at once, each is valued, and the values are summed to a single net amount owed one way or the other. Without it, an insolvency administrator could do what the market calls *cherry-picking*: enforce the contracts that favor the estate and disclaim the rest. Example 1.1 shows what that difference is worth.

Example 1.1 Why the single agreement clause exists

Bank A has five outstanding trades with counterparty B. Marked to market (valued at current market prices) from A's perspective:

$$+40, \quad -35, \quad +20, \quad -15, \quad -8 \qquad (\$ \text{ million})$$

The positives are amounts B owes A; the negatives are amounts A owes B. A's gross positive exposure is $60 million; the net is $2 million in A's favor.

With close-out netting, B's default terminates all five. They sum to $2 million, which A claims from the estate. A's loss is at most $2 million, and if collateral has been posted under the arrangement described below, close to nothing.

Without close-out netting, the administrator enforces the three contracts on which A owes money and disclaims the two on which A is owed. A must pay 35 + 15 + 8 = $58 million in full, and joins the queue of unsecured creditors for its $60 million of claims. At a 20% recovery rate (the share of its claims an unsecured creditor eventually collects), A collects $12 million:

$$\text{A's net payment} = 58 - 12 = \$46 \text{ million, in place of a } \$2 \text{ million claim}$$

The trades did not change. The document did. This is why the enforceability of close-out netting in a given jurisdiction is a precondition for a deep derivatives market there, and why regulators permit netted exposures to be used in capital calculations only where that enforceability has been established.

The second institution is the Credit Support Annex (CSA), an optional schedule to the Master Agreement under which the parties post collateral against the net amount as it moves. The CSA converts a credit exposure into a daily cash flow, which largely removes default risk and puts liquidity risk, the risk of having to find cash at short notice, in its place. That substitution is the subject of Chapter 8, because the second risk turned out to be the one that broke firms in 2022. Together the two documents explain the structure of the numbers in the next section: gross market value is what the contracts are worth, netting under the Master Agreement collapses it to a fraction, and collateral under the CSA covers most of what remains.

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1.6 HOW BIG IS THE MARKET?

Derivatives statistics are quoted in numbers so large that they are routinely misused. The Bank for International Settlements (BIS) puts the notional amount outstanding of over-the-counter derivatives at roughly $850 trillion in recent years ($845 trillion at the end of 2025), something like seven times world gross domestic product (GDP). The figure is correct and almost entirely uninformative.

Notional is the reference amount used to *calculate* payments. It is not an amount anyone owes, expects to receive, or can lose. On a $100 million ten-year interest rate swap, the $100 million is never paid by anyone: it is multiplied by a rate difference to determine a payment of a few hundred thousand dollars. Two more meaningful measures follow.

Gross market value is what the outstanding contracts are actually worth: the sum of what every position with a positive value would be paid if all contracts were settled today. Across the market this runs at roughly $23 trillion, under 3% of notional.

Gross credit exposure nets each pair of counterparties' offsetting positions against one another, as their master agreements permit them to do. That reduces the figure again, to roughly $3.4 trillion, under half a percent of the headline. Most of that remainder is collateralized, by the mechanisms of Chapter 8.

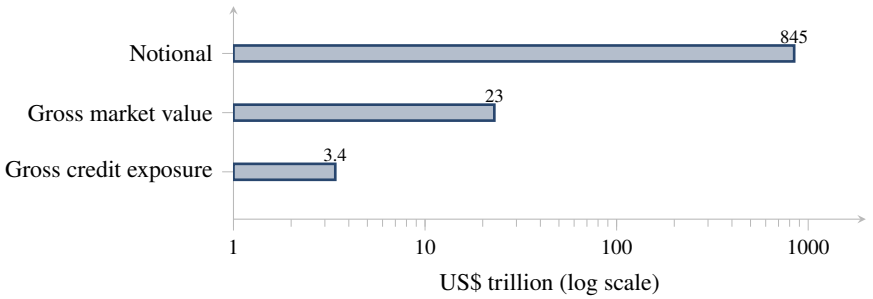


Figure 1.3: OTC derivatives at the end of 2025 (BIS). Log scale.

Figure 1.3 needs a log scale to show all three bars at once, which is itself the lesson: each step down is roughly an order of magnitude. A commentator who reports the first number and implies the third has overstated the risk by a factor of some two hundred and fifty. Example 1.2 traces the same three measures through a single swap.

Example 1.2 What $100 million of notional actually risks

A bank enters a five-year interest rate swap on $100 million notional, paying 4.00% fixed annually and receiving the floating rate.

The notional is never exchanged. In a year when the average floating rate sets at 4.60%, the bank receives

$$100{,}000{,}000 \times (0.0460 - 0.0400) = \$600{,}000$$

and in a year when floating sets at 3.50% it pays $500,000. The annual cash flows are measured in hundreds of thousands on a $100 million "position."

Now suppose rates move sharply and the swap's market value rises to $4 million in the bank's favor. That $4 million—not $100 million—is what the bank would lose if the counterparty failed and nothing had been collateralized. If the same counterparty also owes the bank $1.5 million on one swap and is owed $3 million on another, netting under the master agreement reduces the exposure again, to $2.5 million.

So the same trade supports three "sizes": $100 million of notional, $4 million of market value, $2.5 million of net exposure, which after collateral very likely falls to a few hundred thousand.

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Each is the right answer to a different question, and confusing them is the single most common error in commentary about this market.

Case Study 1.1 Why the headline number will not go away

Notional outstanding is a poor measure of risk, and everyone who works in the market knows it. It nevertheless remains the number that gets quoted, for two reasons worth understanding.

The first is that it is the only figure that can be reported consistently. Market value depends on the valuation date and on models; notional is written into the contract. The second is that notional is not *meaningless*: it scales operational load, and it is the base on which several regulatory measures are calculated. Two portfolios with the same market value but very different notionals are not equally easy to manage, close out, or transfer.

Whenever a derivatives figure is quoted, ask which of the three it is. If the source does not say, it is almost certainly notional, and almost certainly being used to suggest something the number does not support.

What the market is made of

The composition is as lopsided as the size is large. Figure 1.4 gives OTC notional by asset class at the end of 2025.

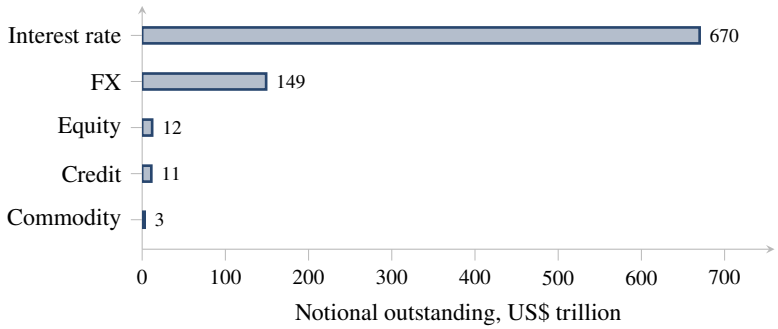


Figure 1.4: OTC notional by asset class, end of 2025 (BIS).

Interest rate contracts are roughly 80% of the total and foreign exchange most of the rest. Credit, equity, and commodity derivatives together are a rounding error on notional—which is precisely why notional misleads, since credit derivatives were at the center of a financial crisis while representing about 1% of it. This lopsidedness explains the shape of this book: Chapters 4 through 9 and 28 through 33 concern rates, because that is where the market is.

Exchange-traded volumes tell a different and more recent story. In 2024 the Futures Industry Association counted a record 206.5 billion futures and options contracts traded worldwide, and the largest single venue was India's National Stock Exchange. In 2025 the count fell 42% to 119.3 billion, most of the fall in equity contracts, though Asia–Pacific exchanges still accounted for well over half of the total. Contract counts are not comparable to notional (an Indian index option and a short-term interest rate futures contract on $1 million differ in size by orders of magnitude), and a swing of that size in a single year is a warning against reading a trend off them. Measured by *activity*, the center of gravity in listed derivatives has nonetheless moved toward Asia, and toward small-size, short-dated, retail-accessible contracts. Chapter 10 returns to this, including the growth of options expiring the same day they are traded, which now account for a large share of US index option volume.

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1.7 WHAT CAN BE AN UNDERLYING

The five asset classes of Figure 1.4 account for almost all the outstanding notional, but they do not exhaust what a derivative can reference, and the exceptions are instructive about why derivative markets succeed or fail.

Weather. The Chicago Mercantile Exchange (CME) lists futures and options on heating and cooling **degree days** for major cities, settling against temperature data published by a national meteorological service. The index is the number of degrees by which each day's average temperature falls below (heating) or rises above (cooling) 65°F, summed over a month or season. An energy utility's revenue falls in a mild winter; a weather contract pays when the winter is mild. The market is small but it works, because it meets the first condition for any derivative market: the settlement index is public, verifiable, and impossible for either party to influence.

Freight. Forward freight agreements, which fix the cost of carrying a cargo on a given route, settle against published shipping route indices. Their users, ship owners with capacity to fill and charterers with cargo to move, have genuinely opposite exposures, which is the second condition for a market to function.

Emissions. Futures on European Union allowances (permits to emit one metric ton of carbon dioxide under the EU's emissions trading system) are now among the more liquid commodity contracts in Europe. Here the underlying exists only because policy created it, which makes the contract's value dependent on the durability of the policy—a risk with no analog in a wheat contract.

Volatility. A derivative can reference the market's expectation of future variability rather than any price. The Cboe Volatility Index (VIX) measures the volatility that S&P 500 option prices imply for the next 30 days; VIX futures, listed in 2004, and options on them made volatility an asset class in its own right; Part IV is devoted to it.

Cryptoassets. CME listed bitcoin futures in December 2017 and ether futures subsequently. After US spot bitcoin exchange-traded funds (ETFs) were approved in January 2024, listed options on those funds followed later that year, adding options on a spot product to the options on bitcoin futures that CME had listed since 2020. The instruments are entirely conventional; what is novel is the underlying and the extreme volatility that pricing must accommodate.

Other derivatives. An option on a futures contract, or a swaption (an option to enter a swap), references a derivative rather than an asset. These are not curiosities: swaptions are among the largest option markets in the world, and Chapter 29 prices them.

What makes a derivative market possible. The list suggests three conditions, and they explain the failures as well as the successes. There must be an *observable, verifiable settlement reference* that neither party can manipulate. There must be participants with *genuinely opposite exposures*, so that trade is risk transfer rather than a wager between two people who both merely have opinions. And the exposure must be *material enough to be worth managing*, since a contract no one needs will not attract the liquidity that makes it usable.

Proposals for derivatives on macroeconomic aggregates (GDP growth, inflation-adjusted national income, house price indices for individual homeowners) have repeatedly satisfied the first condition and failed the second. Almost everyone is exposed to a recession in the same direction, so there is no natural party to take the other side, and the contracts have not sustained liquidity despite decades of advocacy. When a proposed contract seems obviously useful and does not exist, the missing party is usually the explanation.

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1.8 WHO USES DERIVATIVES, AND WHAT FOR

Textbooks traditionally divide participants into hedgers, speculators, and arbitrageurs. The taxonomy is useful and Section 1.9 takes it seriously, but on its own it tells a reader almost nothing about who is actually on the other side of a trade. Table 1.1 is more concrete: real institution types, the exposure each has, and the instrument each reaches for.

Table 1.1: Who trades what, and against which exposure.

Participant	Exposure	Typical instrument
Airlines, shippers	Fuel cost	Crude and refined product swaps, options
Exporters, importers	Receivables and payables in foreign currency	FX forwards, currency options
Corporate borrowers	Floating-rate debt service	Pay-fixed swaps, interest rate caps
Bond issuers	Locking a coupon before issue	Forward-starting swaps, swaptions
Pension funds, insurers	Long-dated liabilities discounted at market rates	Receive-fixed swaps, gilt and Treasury futures
Asset managers	Portfolio duration and equity beta	Bond and index futures
Banks (loan books)	Concentrated credit exposure	Credit default swaps, index tranches
Dealers	Inventory acquired from clients	Whatever offsets it, plus the futures strip
Hedge funds	Deliberately acquired relative-value positions	Everything in this book

Two observations about the table matter more than its contents. First, most of these participants are *removing* an exposure they acquired by doing something else: flying aircraft, writing insurance, lending money. Derivatives markets exist mainly to serve them, and the profits earned by the other side are, in the aggregate, the price those participants pay for the service. Example 1.3 follows one of them through a quarter.

Second, the dealer row is different in kind. A dealer does not begin with an exposure; it acquires one by accommodating clients, and then manages it. That is a business of intermediation, and understanding it explains a great deal that is otherwise puzzling about market prices: why a heavily demanded contract trades rich, why hedging costs rise when everyone hedges at once. This book returns to it repeatedly, and Chapters 8 and 9 are about what that intermediation costs.

Example 1.3 An airline hedging fuel

An airline expects to burn 12 million gallons of jet fuel next quarter. Fuel is roughly 25% of its operating cost, and a 30-cent move in the price per gallon changes quarterly fuel spend by $3.6 million, enough to swing a thin operating margin to a loss.

There is no liquid jet fuel futures contract, so the airline hedges with heating oil futures, whose underlying is a similar refined product. Each contract covers 42,000 gallons. Jet fuel has historically moved about 0.64 of a cent for each cent of heating oil (Example 3.3 estimates that ratio), so the airline buys

$$\frac{12{,}000{,}000}{42{,}000} \times 0.64 = 182.9 \rightarrow 183 \text{ contracts}$$

Suppose jet fuel then rises 30 cents a gallon and heating oil futures rise 34 cents. The airline pays $3.6 million more for fuel and gains

$$183 \times 42{,}000 \times 0.34 = \$2{,}613{,}240$$

on the hedge, leaving about $0.99 million of the increase uncovered. Jet fuel rose 30 cents for

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every 34 of heating oil, a ratio of 0.88 against the 0.64 the hedge was sized on, so in this quarter the hedge was too small. The ratio is set on the average relationship, and in any one quarter the realized ratio can sit well above or below it.

Three things in this example run through the whole book. The hedge is *approximate*, because the traded instrument is not the exposure (the cross-hedging problem of Section 3.5). The residual is not small relative to the margin being protected. And the gain on the futures leg arrives as cash, daily, whether or not the fuel has yet been bought—which is a benefit here and, in Chapter 3's failure cases, a catastrophe.

1.9 HEDGING, SPECULATION, AND ARBITRAGE—AND WHY THE LABELS BLUR

With the participants in view, the traditional triad can be stated properly. A **hedger** holds an existing exposure and enters a derivative that offsets it. The test is not the trade but the portfolio: the same short futures position is a hedge for a producer and a directional bet for someone with no production.

A **speculator** takes a position to profit from a view. Speculators are frequently described apologetically, as though merely tolerated, and that is a mistake. Hedgers rarely want to shed opposite exposures in the same amount at the same moment, and someone must take the imbalance; a market of hedgers alone would seldom clear. What speculation supplies is the willingness to hold an exposure no one else wants, and the price of that service is a real cost to hedgers, as Section 5.14 shows when it works out who pays whom.

An **arbitrageur** exploits a price inconsistency between related instruments, in principle without taking risk. This is the category that requires the most caution, and Section 5.15 is devoted to why: the textbook arbitrage assumes unlimited borrowing at a single risk-free rate, and the real trade requires a balance sheet, financing that may be withdrawn, and the ability to survive the position moving against you before it converges. Covered interest parity, the rule that a forward exchange rate equals the spot rate adjusted for the two currencies' interest rates, is violated persistently in the deepest currency markets in the world, and the violations exist because that assumption is false. Example 1.4 applies all three labels to a single trade.

Example 1.4 One trade, three labels

A trader buys 100 Treasury bond futures contracts. Which of the three is it?

If the trader runs a pension fund whose liabilities are longer-dated than its bonds, so that a fall in yields would raise the liabilities by more than the assets, it is a hedge: the futures offset an exposure the fund already carries.

If the trader believes long yields will fall and holds nothing else, it is speculation.

If the trader has computed that the futures are cheap after carry (the bond's coupon income less the cost of financing it) relative to the cheapest-to-deliver cash bond (the bond a futures seller would choose to deliver), and simultaneously sells that bond short, borrowing it through a reverse repo (lending cash against the bond as collateral; Section 4.1 describes the repo market), it is arbitrage: the basis trade of Section 6.5, run in reverse.

The three are the same instrument, the same direction, and the same quantity. Only the rest of the balance sheet distinguishes them, which is why regulators, risk managers, and accountants all find these categories so difficult to apply, and why a firm's own description of a position is weak evidence about what it is. Chapter 3 describes a firm whose "hedge" was economically sound and failed anyway, and Section 5.14 describes a fund whose "spread trade" carried more risk than an outright position would have.

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1.10 LEVERAGE

The feature that makes derivatives efficient is the same one that makes them dangerous, and it is worth isolating. To gain $10 million of exposure to the S&P 500 by buying shares requires $10 million. To gain the same exposure through index futures requires posting initial margin, a good-faith deposit sized to cover a few days of adverse moves, and returned when the position closes. If that margin is 5% of notional, the same exposure costs $500,000 to establish. Example 1.5 sets this route beside three others.

Table 1.2: $10 million of index exposure, four ways.

Route	Cash required	Loss on 10% fall	% of cash
Buy the shares	$10,000,000	$1,000,000	10%
Buy on 50% margin	$5,000,000	$1,000,000	20%
Index futures (5% IM)	$500,000	$1,000,000	200%
At-the-money call options	$600,000	$600,000	100%

Example 1.5 The same exposure, four ways

A fund wants $10 million of exposure to an equity index. Table 1.2 shows what each route requires and what a 10% market fall does to the capital committed.

The futures row is the one to study. A 10% fall produces a loss of twice the cash committed. The position is not merely wiped out—the fund owes $500,000 more, and will be asked for it in cash the following morning as variation margin, the daily settlement of gains and losses that Section 1.1 describes. Initial margin is not the maximum loss; it is a deposit, and Chapter 2 explains what happens when it is exhausted.

The option row (calls whose strike equals the current index level, which is what *at the money* means) differs in kind rather than degree. The premium is the maximum loss in every state of the world, which is why Section 12.10 argues that buying options is sometimes the right answer even when it looks expensive. The cost of that protection is the $600,000 premium, which is lost entirely if the index is unchanged.

Table 1.2 also explains a structural fact about the market: derivatives let a position be built that no lender would finance directly. A firm that wanted $10 billion of equity exposure would find it difficult to borrow $10 billion; it can obtain the same exposure through total return swaps with several dealers, none of whom sees the whole picture. Whether the resulting position is prudent is a separate question from whether anyone can observe it, and Archegos Capital Management showed in March 2021 what happens when the answer to both is no. It built concentrated positions in a few US and Chinese stocks almost entirely through total return swaps with several prime brokers, the banks that finance and execute trades for hedge funds and similar clients. The banks owned the shares, so Archegos filed no ownership disclosures, and each broker saw only its own exposure, with no way to know that competitors had written substantially the same trade against the same client. When the stocks fell, margin calls arrived at every broker at once, Archegos could not meet them, and the banks' simultaneous liquidations drove prices further down; the banks lost more than $10 billion between them. Section 27.14 works the numbers and Section 40.6 tells the episode.

1.11 WHAT GOES WRONG

Derivatives have a reputation for producing spectacular losses, and it is deserved. What is less well understood is that the losses follow a small number of repeated patterns. Table 1.3 lists the episodes this book returns to, and Chapter 40 takes them up together.

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Table 1.3: Episodes and the mechanism each illustrates.

Episode	Year	Mechanism
Metallgesellschaft	1993	A long-dated hedge sound in value but funded with short-dated futures; rolls and margin calls exhausted liquidity first (Chs. 3, 40)
Barings	1995	Unsupervised trader, unreconciled accounts; the loss was in plain futures and options (Ch. 40)
Long-Term Capital Management	1998	Convergence trades that were right eventually, financed in a way that could not survive being wrong first (Ch. 40)
Amaranth	2006	Calendar spreads with low margin and no exit; size itself was the risk (Chs. 5, 40)
Global financial crisis	2007–09	Credit derivatives on correlated collateral, valued with models calibrated on a benign sample (Chs. 24–26)
Negative WTI settlement	2020	Retail products whose design assumed prices could not go below zero (Chs. 5, 36)
Archegos	2021	Concentrated leverage invisible across dealers (Chs. 27, 40)
LME nickel	2022	A short-covering spike that broke the clearing house's own rules (Chs. 2, 40)
European energy	2022	Solvent hedgers unable to fund variation margin (Chs. 8, 40)
UK pension LDI	2022	Leveraged rate hedges, held in liability-driven investment mandates, liquidated into the market they were hedging against (Chs. 8, 40)

Read down the mechanism column and a pattern emerges that is not the expected one. Very few of these were valuation failures. The models mostly worked. What failed was *funding*: a position that was economically correct required cash at a moment when cash was unavailable, and had to be closed at the worst possible time.

This is the book's most persistent theme. A hedge that is right about price can still fail through liquidity. That sentence would have been an odd thing to build a derivatives textbook around in 2005. After 2008, 2020, and 2022, it is the central practical lesson of the subject, and it is why Chapter 8 (collateral, margin, and funding) exists as a chapter in its own right rather than as an appendix to the swaps chapter.

The second pattern is that scale and invisibility travel together. Barings, Amaranth, and Archegos each involved a position that had grown too large to exit, and in each case those who could have stopped it never saw its full size: senior management at Barings, the New York Mercantile Exchange, which saw only the part of Amaranth's position held there, and the dealers facing Archegos, each of which saw only its own slice.

1.12 REGULATION SINCE 2008

The market described in this book is substantially a creation of policy decisions taken after 2008, and a reader who learned the pre-crisis structure would find much of current practice inexplicable. At the Pittsburgh summit in September 2009, the G20 agreed four commitments for OTC derivatives:

- standardized contracts should be centrally cleared;
- they should trade on exchanges or electronic platforms where appropriate;
- all contracts should be reported to trade repositories;
- non-centrally-cleared contracts should attract higher capital.

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Implementation took over a decade and reshaped the market:

Clearing mandates moved the great majority of interest rate and index credit derivatives to CCPs, which is why Chapter 2 treats clearing houses as systemically important institutions rather than as infrastructure.

Trade reporting created the data underlying the statistics of Section 1.6, and, less happily, a compliance burden that consolidated dealing into fewer firms.

Uncleared margin rules, phased in from 2016 and completed in September 2022, require two-way initial margin on bilateral trades between large participants. This changed pricing, not merely operations: posting initial margin has a funding cost, which is the margin valuation adjustment (MVA) of Chapter 9.

Capital rules (the credit valuation adjustment charge, the standardized approach for counterparty credit risk, and the leverage ratio) made dealer balance sheets an explicit input into derivative prices. Section 5.15's argument that arbitrage requires a balance sheet is in large part an argument about these rules.

Who writes the capital rules is worth stating once, because every later chapter assumes it. The **Basel Committee on Banking Supervision**, a body of central banks and supervisors hosted at the Bank for International Settlements, agrees standards (Basel I in 1988, Basel II in 2004, and the post-crisis Basel III), which have no legal force until each jurisdiction writes them into its own rules, as the European Union, the United States, the United Kingdom, Canada, and China separately do. A bank must then hold capital, meaning common equity first and other loss-absorbing instruments after it, of at least 8% of its risk-weighted assets (each exposure multiplied by a weight meant to reflect its riskiness), plus buffers that take the requirement above 10%, and separately at least a few percent of its *unweighted* exposure under the leverage ratio. Everything Chapters 9, 26, 27, and 39 say about capital is an elaboration of those two ratios, Section 27.11 assembles them for a whole bank, and the divergences those chapters report (Europe exempting a charge the United States does not) come from the national step rather than from the accord.

Two developments are more recent and still unfolding. Benchmark reform retired LIBOR in favor of overnight risk-free rates, an event that changed the foundations of interest rate derivative pricing rather than its details; Chapter 4 is written from the new foundation rather than adapting the old one. And clearing continues to expand: the mandate from the Securities and Exchange Commission (SEC) to centrally clear US Treasury cash and repo transactions, adopted in December 2023 and phasing in over the following years, extends the post-crisis model to the largest government bond market in the world. Section 2.11 takes it up, with its compliance dates, and Section 6.5 its effect on the basis trade.

1.13 HOW THE CHAPTERS FIT TOGETHER

The book serves three audiences with different needs, and the shape it takes is the same for all three. Part I (Chapters 1–9) covers markets, forwards, futures, swaps, and the collateral and valuation machinery that modern practice requires. Part II develops options mechanics and strategies, and Part III their valuation, from the binomial tree through Black–Scholes–Merton, the Greeks (the sensitivities of an option's value to its inputs), and the volatility surface. Part IV treats volatility as an asset class. Parts V–IX cover credit, interest rate derivatives in depth, exotics and structured products, commodity and digital markets, and risk and regulation.

Figure 1.5 is the map. Within a part the chapters run in order and each assumes the one before it; the arrows are the principal cross-part dependencies, and they are what a reader assembling a path actually needs.

Three chapters are shaded as hubs, because the rest of the book stands on them. Chapter 4's risk-free-rate conventions are assumed by every rates chapter that follows. Chapter 13's

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binomial tree is assumed by everything in valuation, exotics, and computation. And Chapter 15 is the gate between those and the rest: later chapters open by naming it as a prerequisite almost as often as they name Chapter 13.

A hub is a chapter you must pass through, not one you keep turning back to, and the distinction is worth drawing because the two rankings disagree. By the number of later chapters that cite it at all, the busiest chapter in the book is Chapter 19, whose volatility surface is referred to from most of the chapters that follow it, but almost none of them need it as a prerequisite. Chapter 15 is the reverse: cited less often, and skippable by no one.

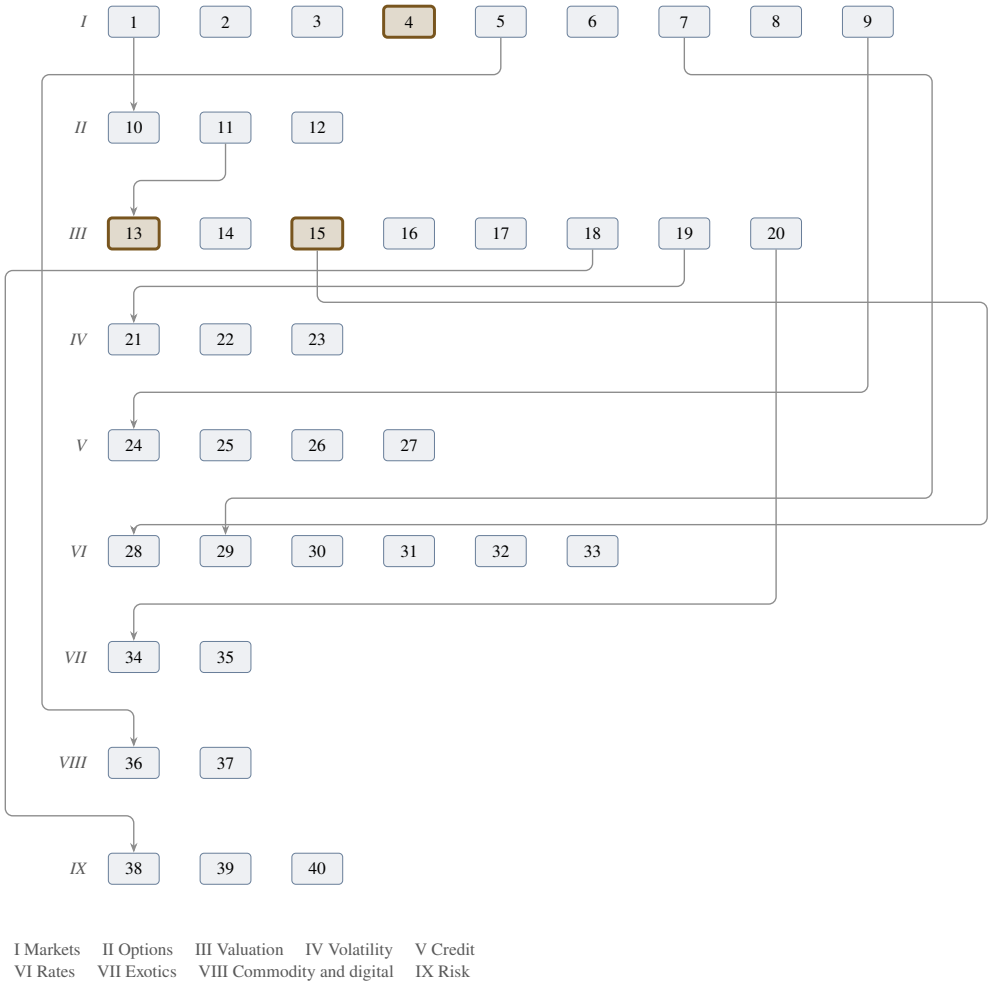


Figure 1.5: Chapters run in order within a part; arrows are the principal cross-part dependencies. Shaded chapters are the three hubs.

SUMMARY

A derivative is a contract whose value depends on something else, and its economic function is to separate exposure to a risk from ownership of the thing that carries it.

Four building blocks account for nearly everything: forwards, futures, swaps, and options. The first three are symmetric (both parties are obliged and the payoff is linear), while options are asymmetric, giving one party a choice and requiring a premium in exchange. The distinction

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organizes the whole subject.

Markets are exchange-traded, bilateral OTC, or—increasingly—bilaterally negotiated and centrally cleared. The third category is a post-2009 creation and now holds most of the outstanding interest rate risk in the world.

The market's size is routinely misstated. Notional outstanding is the figure usually quoted, but gross market value is a small fraction of it and exposure after netting a smaller fraction still, most of it collateralized. Interest rate contracts make up the large majority of the notional total, which is why this book weights them heavily.

Participants are better described by what they do than by the hedger–speculator–arbitrageur triad, though the triad remains useful once it is understood that the same trade falls into different categories depending on the rest of the balance sheet. Dealers are a distinct case: they do not begin with an exposure but acquire one by serving clients, and the cost of that intermediation shows up in prices.

Leverage is the source of both the efficiency and the danger. Initial margin is a deposit, not a maximum loss; a futures position can lose several times the cash committed, while a purchased option cannot lose more than its premium.

Most derivative disasters were not valuation failures. They were funding failures: positions that were economically correct but required cash when none was available. That theme recurs throughout the book and justifies the prominence given to collateral and margin.

FURTHER READING

The BIS publishes semiannual OTC derivatives statistics and quarterly exchange-traded data; they are the authoritative source for Section 1.6 and are worth consulting directly, since the definitions matter as much as the numbers. The Futures Industry Association's annual volume survey covers listed markets. For the post-crisis reform program, the Financial Stability Board's annual progress reports on OTC derivatives reform track what was promised against what was delivered. On Archegos, the Credit Suisse board's external investigation report sets out how the exposure went unmeasured.

PRACTICE QUESTIONS

1.1 Explain why a forward contract has zero value at inception but an option does not, and what this implies about when money changes hands in each.

1.2 A company enters a \$50 million notional five-year swap paying 3.8% fixed and receiving floating. In the first year floating averages 5.1%. Compute the net payment and state who pays whom. What fraction of notional is it?

1.3 Distinguish notional amount, gross market value, and gross credit exposure. A newspaper reports that a bank "has \$3 trillion of derivatives." What further question should a reader ask?

1.4 An investor obtains \$4 million of index exposure through futures requiring 6% initial margin. The index falls 12%. Compute the loss, express it as a multiple of the initial margin, and state what the investor owes.

1.5 Explain why a market consisting only of hedgers would not function, and identify what speculators are being paid for.

1.6 Classify each as hedging, speculation, or arbitrage, and state what additional information you would need in each case: (a) a copper producer selling copper futures; (b) a fund buying gold futures and selling gold forward at a higher price; (c) an importer buying dollars forward; (d) a bank buying protection on a borrower to whom it has lent.

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1.7 Using Figure 1.1, explain why the writer of a call is in a different business from the buyer of one, and what the writer must be able to do to be in it safely.

1.8 Explain what changes when the same economic trade is executed on an exchange rather than bilaterally. Identify which of those changes transfers risk to someone else and which merely changes when cash moves, and state one exposure that the exchange does not remove.

1.9 Using Example 1.1, explain what the single agreement clause prevents an administrator from doing in a bankruptcy, and use it to account for the fall from $23 trillion of gross market value to $3.4 trillion of gross credit exposure.

1.10 Archegos, as this chapter describes it, held a position no counterparty could see in full. Identify the instrument that made it invisible, state what each prime broker could observe and what it could not, and explain what would have had to change (in the instrument, the reporting, or both) for the aggregate position to be visible to any one of them.

FURTHER QUESTIONS

1.11 Table 1.3 attributes most of the listed episodes to funding rather than valuation failure. Choose two episodes and explain precisely what cash was required, when, and why it was not available. Then state what a risk report would have had to measure to give warning.

1.12 Central clearing was intended to reduce systemic risk by replacing a web of bilateral exposures with a hub. Construct an argument that it instead concentrates risk, and identify what would have to be true for the concentration to be worth accepting. Chapter 2 gives the full treatment; attempt it first.

1.13 In the Archegos episode, each dealer's individual exposure was defensible. Explain how a set of individually prudent positions produced a collectively imprudent one, and evaluate whether a disclosure rule attaching to economic exposure rather than share ownership would have prevented it.

1.14 An airline hedges fuel and the fuel price then falls, so the hedge loses money and the airline's unhedged competitor reports lower costs. Construct the argument that the hedging program should be continued, and the argument that it should not. Which turns on facts about the airline rather than about fuel?

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